Joseph A. Smith

Environmental Systems & Intelligence Specialist | Infrastructure | Applied AI & Geospatial Systems
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TARGET

Environmental systems and operations professional: wastewater and pump systems, environmental sampling and compliance, mechanical/electrical troubleshooting, GIS and evidence-based reporting.

PROFESSIONAL PROFILE

Self-directed environmental and infrastructure professional with hands-on experience in wastewater systems, environmental compliance, mechanical systems, construction, field operations and industrial work — now applying that ground-level knowledge to environmental intelligence, geospatial data and AI-assisted system design. Built NviroGear through independent study and practical experimentation rather than a traditional software or engineering degree. Known for persistent problem-solving, learning unfamiliar systems quickly, and connecting physical infrastructure, environmental evidence and digital tools into useful workflows.

PROFESSIONAL EXPERIENCE

Fleis & VandenBrink (F&V;) — 2.5 years

Environmental / Wastewater Field Technician

- Conducted environmental sampling and wastewater system monitoring for regulatory compliance; recorded flow, meter and equipment-performance data.
- Inspected lift stations, pumps and wastewater infrastructure; assisted with diagnosis of mechanical and electrical failures and produced field/compliance documentation.
- Field work included the Zephyr remediation site in Muskegon — a multi-million-dollar cleanup program tied to the Muskegon Lake Area of Concern, where documentation discipline is the job.

NviroGear LLC — Recent

Founder / Environmental Intelligence & Systems Designer

- Conceptualized and built environmental intelligence workflows that combine public environmental records, parcel information, GIS layers, groundwater/well data and contamination evidence into property- and site-level decision support.
- Designed the evolving Alice land/environmental intelligence system and related modular workspaces, focusing on traceable evidence, canonical records, map layers, public/private information controls and reusable application components.
- Developed practical data-ingestion and normalization workflows for county parcel records, environmental datasets, well records and regulatory information; worked with JSON/GeoJSON, database schemas, APIs, web applications and cloud deployment tools.
- Applied AI as a build and research partner to prototype software, investigate unfamiliar technical problems, document systems and turn environmental/field knowledge into usable digital products — while continuing to learn the underlying technologies.
- Explored groundwater-gradient, contamination-proximity and property-intelligence methods to help translate complex environmental records into understandable reports and interactive mapping experiences.
- Directed product and systems concepts spanning environmental compliance workflows, evidence management, modular software architecture, inventory/service systems and automation.

Arborist & Tree Climbing Services — 5+ Years

Owner / Operator

- Operated an independent tree-service business, managing customers, job planning, equipment, removals, trimming, climbing and high-risk rigging under strict safety procedures.

ADDITIONAL INDUSTRIAL & INFRASTRUCTURE EXPERIENCE

Diesel Mechanic / Heavy Equipment Technician

- Rebuilt semi-truck diesel engines, diagnosed failures and performed major repair and preventive maintenance.

Lead Framer — Commercial Construction (Walmart Projects)

- Led crews, interpreted architectural blueprints, coordinated structural layouts and operated lifts/equipment.

Material Handler / Manufacturing Support — Wacker Neuson

- Operated forklifts/cranes and managed production materials and inventory using SAP.

CDL Driver — Team Lindy Trucking

- Regional tractor-trailer operation, inspections and safety documentation.

CORE STRENGTHS

Wastewater & Pump Systems · Environmental Sampling & Compliance · GIS / Geospatial Workflows · Environmental Data Architecture · AI-Assisted Research & System Design · Mechanical / Electrical Troubleshooting · Groundwater & Contamination Intelligence · Field Reporting & Evidence Documentation · Construction & Blueprint Reading · Industrial Equipment & Safety · Rapid Self-Directed Technical Learning

CERTIFICATIONS & SAFETY

Completed: OSHA / HAZWOPER 40-Hour · Industrial Stormwater Certification · Underground Storage Tank Inspection Certification · Respiratory Protection / PPE Certification
Status: CDL — eligible for renewal

TECHNICAL & APPLIED SYSTEMS

Pump systems · Diesel engines · Electrical panels · Mechanical diagnostics · Industrial machinery · Environmental monitoring equipment · Blueprint interpretation · GIS / map-layer concepts · Environmental data normalization · Database / schema concepts · APIs · JSON / GeoJSON · AI-assisted prototyping and research · Cloud / web application workflows

INDEPENDENT RESEARCH & TECHNICAL CURIOSITY

Longstanding interest in acoustics, recording and frequency signatures, including independent exploration of how acoustic/frequency behavior can be applied to liquids, separation and sensing concepts. Music/recording experience contributed to this systems-oriented interest. Approaches new technical areas as an active learner: test, document, understand, improve and keep going.

FIELD MAP & PORTFOLIO

Interactive Michigan field-experience map, labeled project cards and full portfolio at **josephsmith.info**